

Bus Booking App Frontend Analysis

- Built with Zone.js + Observables + HttpClient
- No Signals - uses plain class properties instead
- Modern @if and @for templates
- DI with inject() everywhere
- Standalone components, no NgModules

What We Learned vs What AI Built

- Zone.js Change Detection – Used
 - this.trips = response triggers re-render
 - isLoading = true/false refreshes UI
- Signals - Not used
 - No signal(), computed(), effect()
 - Uses manual applyFilters() instead of computed
- HttpClient - Clean services
 - TripService, AuthService, BookingService, etc.
 - JWT interceptor auto-attaches tokens
- Dependency Injection - Correct
 - @Injectable({ providedIn: 'root' })
 - inject() in all components
- @Component & @Injectable - Every file
 - All decorated properly
- Observables - Three patterns
 - HTTP: http.get() + subscribe()
 - Forms: debounceTime + switchMap + distinctUntilChanged
 - Auth: BehaviorSubject broadcasts role
- @for - Used everywhere
 - Correct track expressions
- @if - Loading, roles, empty states
 - One *ngIf left in bus-layout-designer

What's Good

- Clean service-based architecture
- Proper DI - no manual instantiation
- Real RxJS operators - debounce, switchMap, etc.
- Modern template syntax
- BehaviorSubject for state broadcast

What Needs Work

- No Signals - biggest gap, use computed() for filtered lists

- Subscription leaks - no takeUntilDestroyed()
- Manual derived state - applyFilters() called after every change
- Inconsistency - one *ngIf remains

How a Search Works (One Flow)

- User types in search field
- valueChanges Observable fires
- debounceTime(180) waits for typing to stop
- switchMap calls TripService.getSourceSuggestions()
- HttpClient GET /api/routes/sources
- subscribe() updates sourceSuggestions
- Zone.js detects change
- @for renders dropdown
- User clicks SEARCH
- searchTrips() calls service, isLoading = true
- @if (isLoading) shows spinner
- Response arrives, list renders via @for

Next Steps

- Convert trips/filteredTrips to signal() + computed()
- Add takeUntilDestroyed() to subscriptions
- Study home.component autocomplete pipe chain
- Replace *ngIf with @if
- Trace one API call from click to JSON